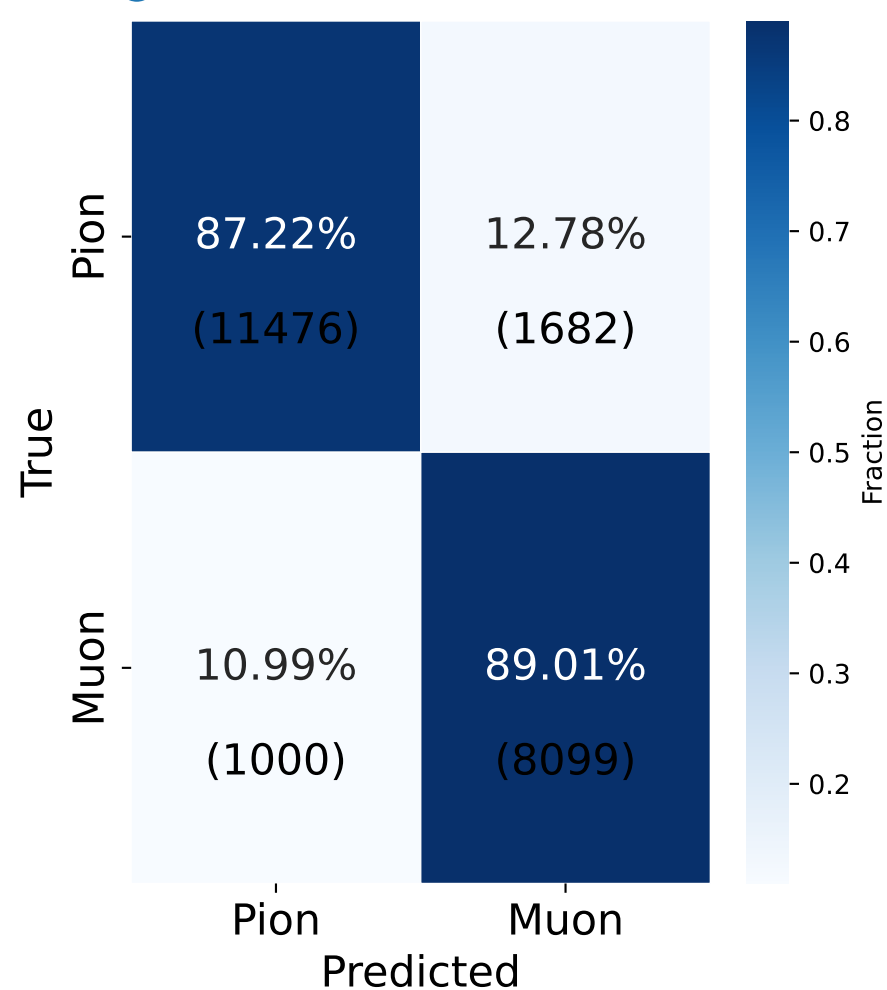


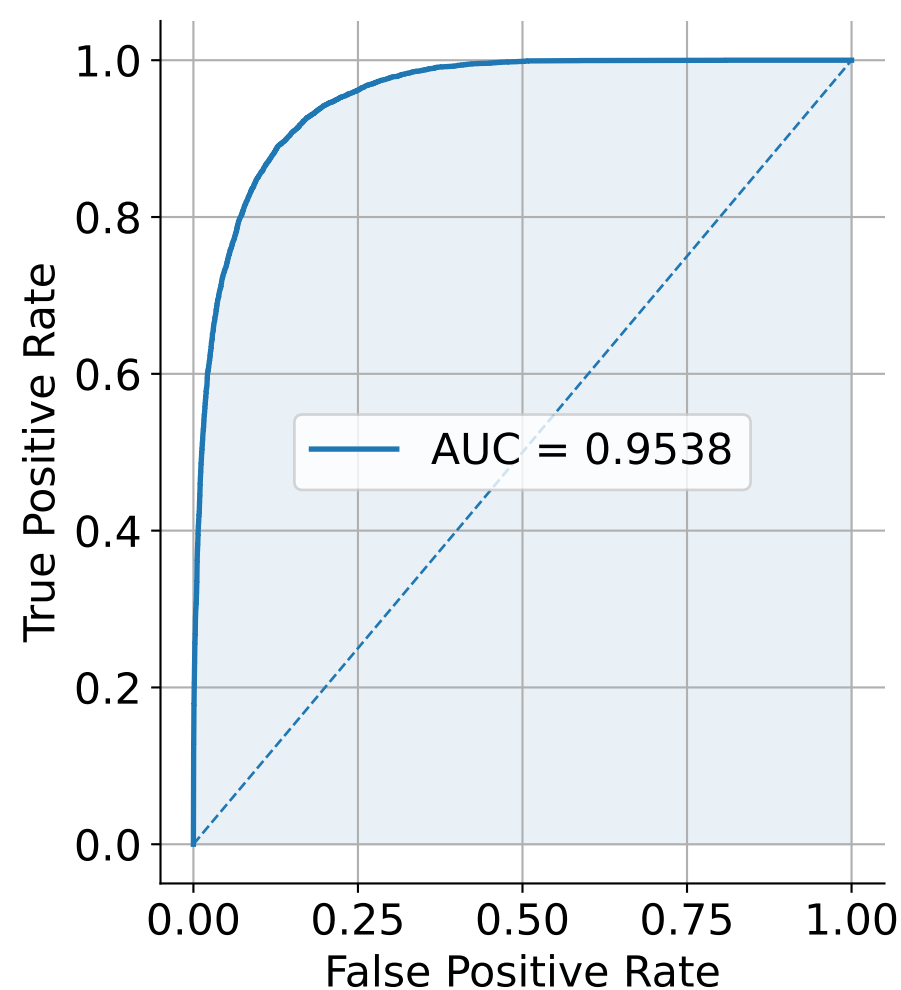
Muon vs Pion ID — XGBoost (Calorimeter + ToF features)

[Train S/B=1:1 | Test S/B=1:1 (realistic, all files)]

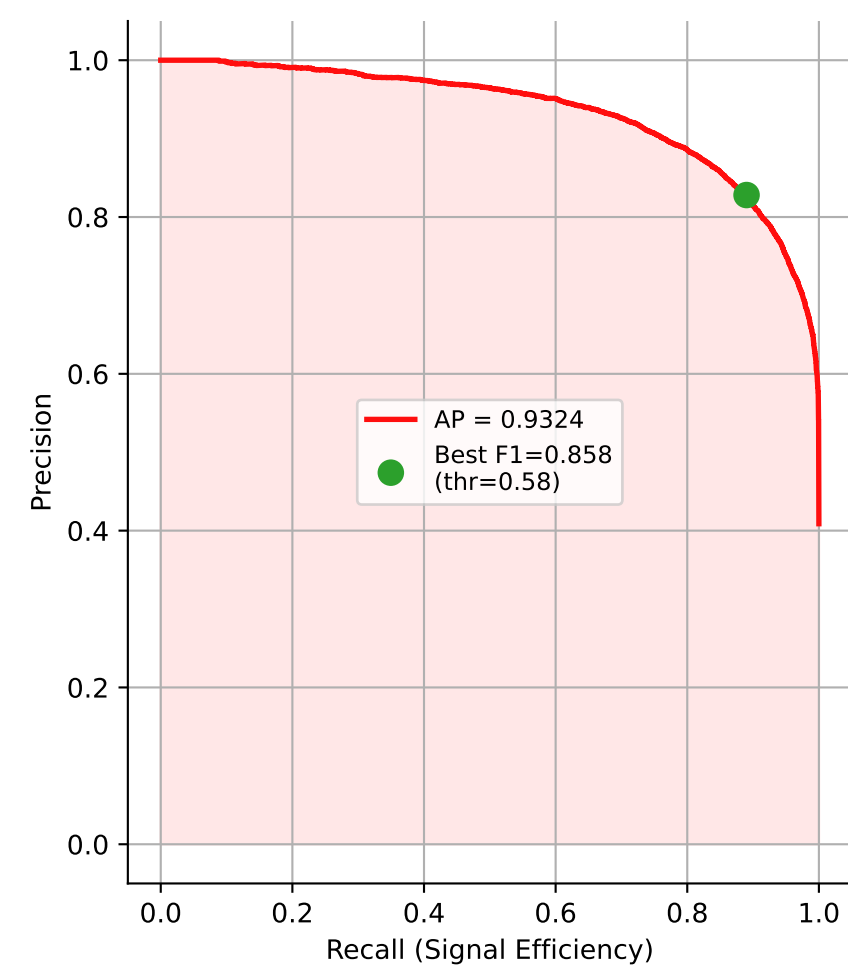
Confusion Matrix
@ Max F1-Score (thr=0.582)



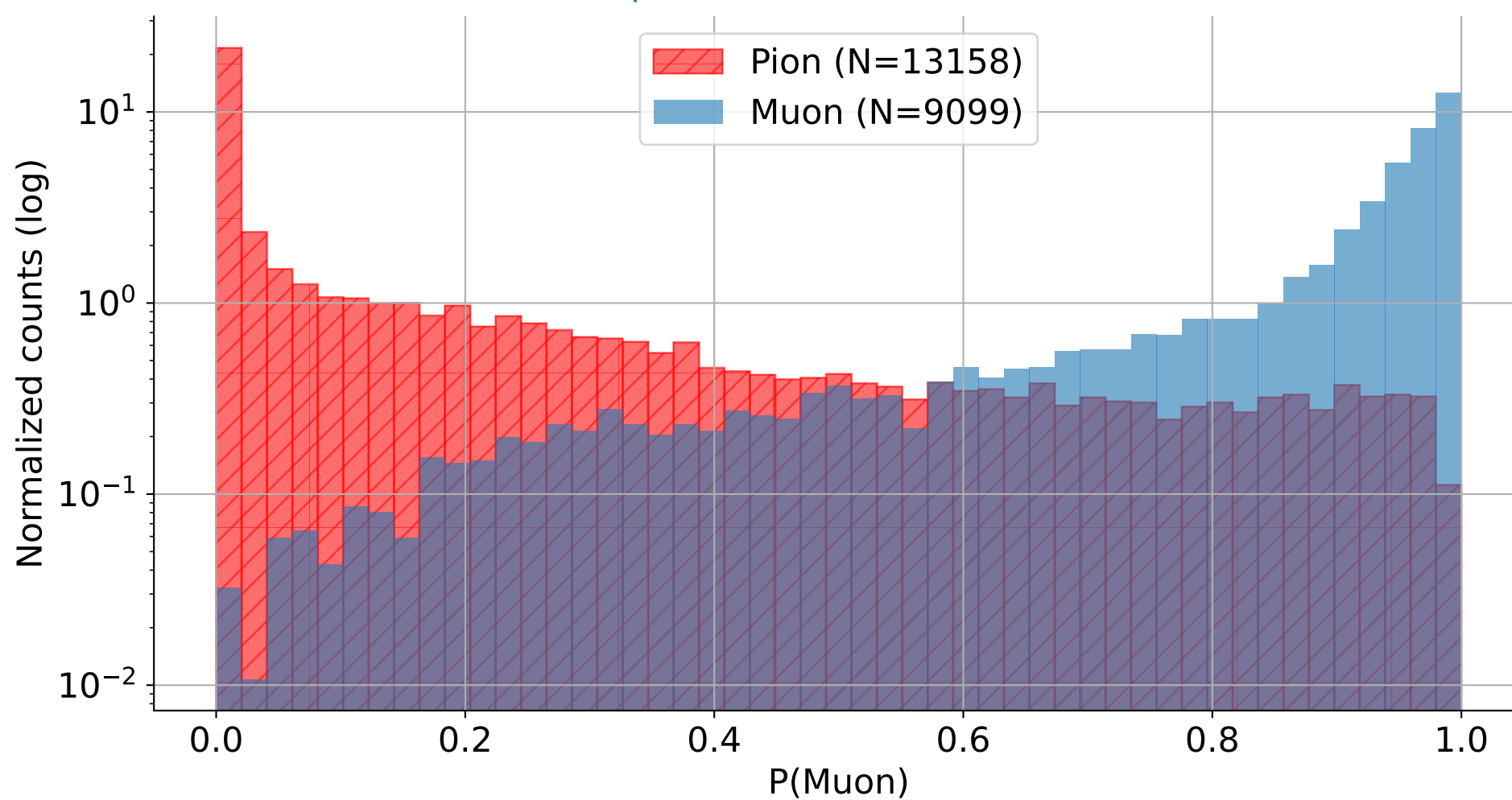
ROC Curve



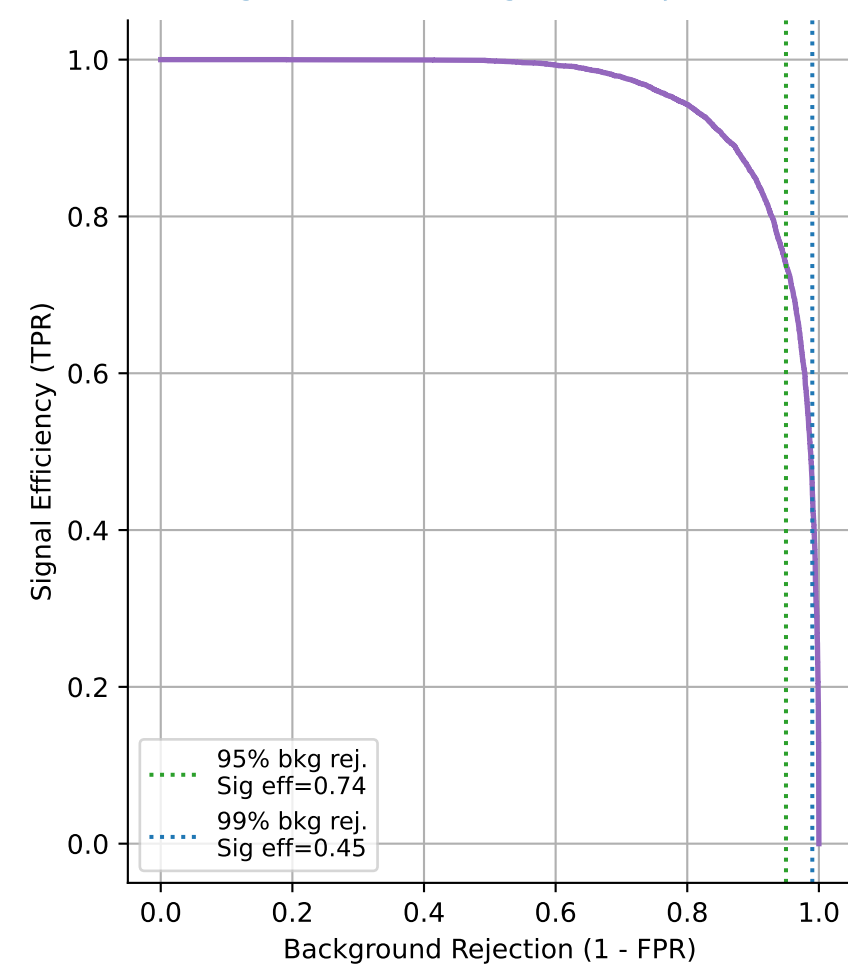
Precision-Recall Curve



Classifier Response Distribution [Test: realistic S/B]

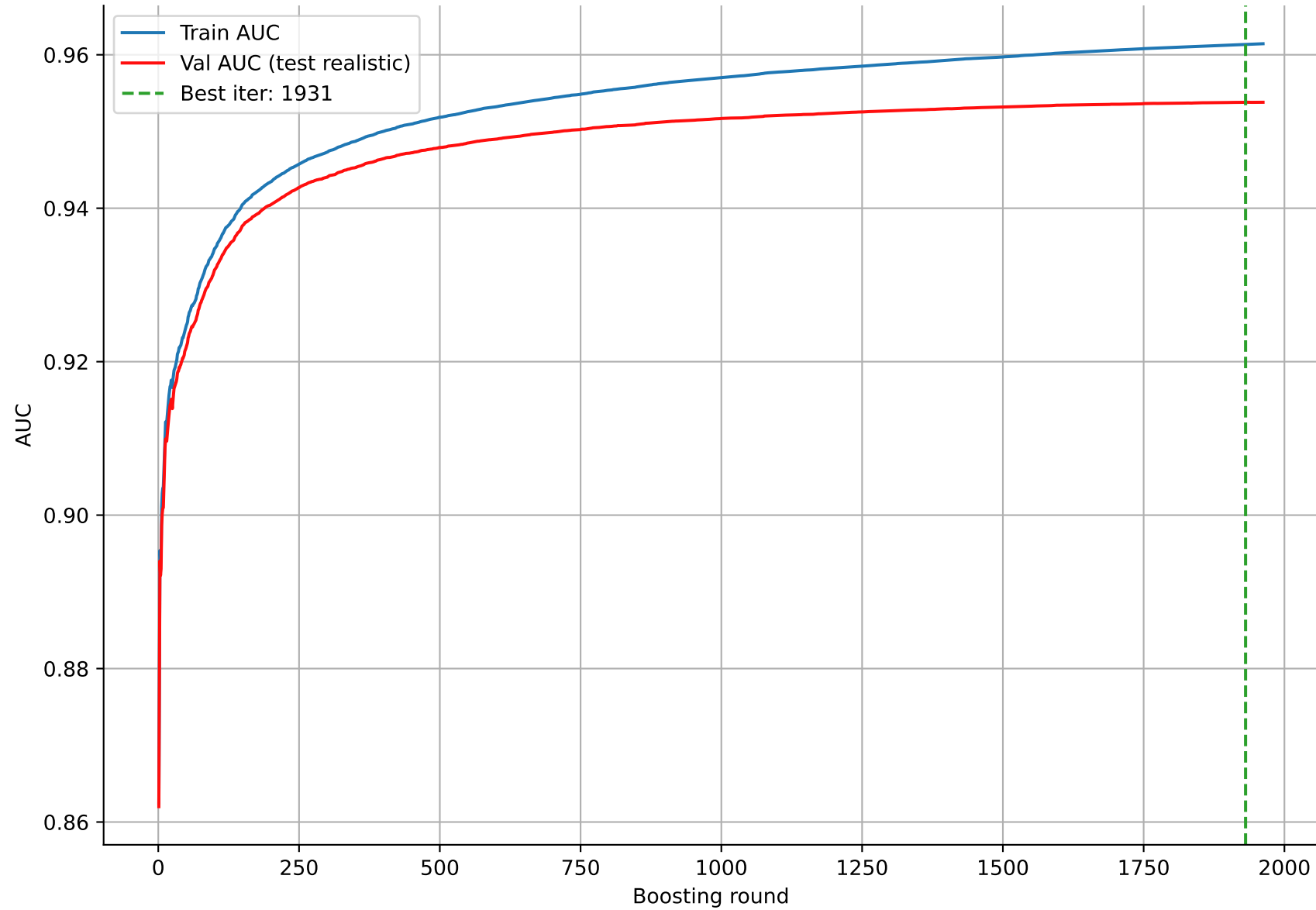


Signal Eff. vs Background Rejection

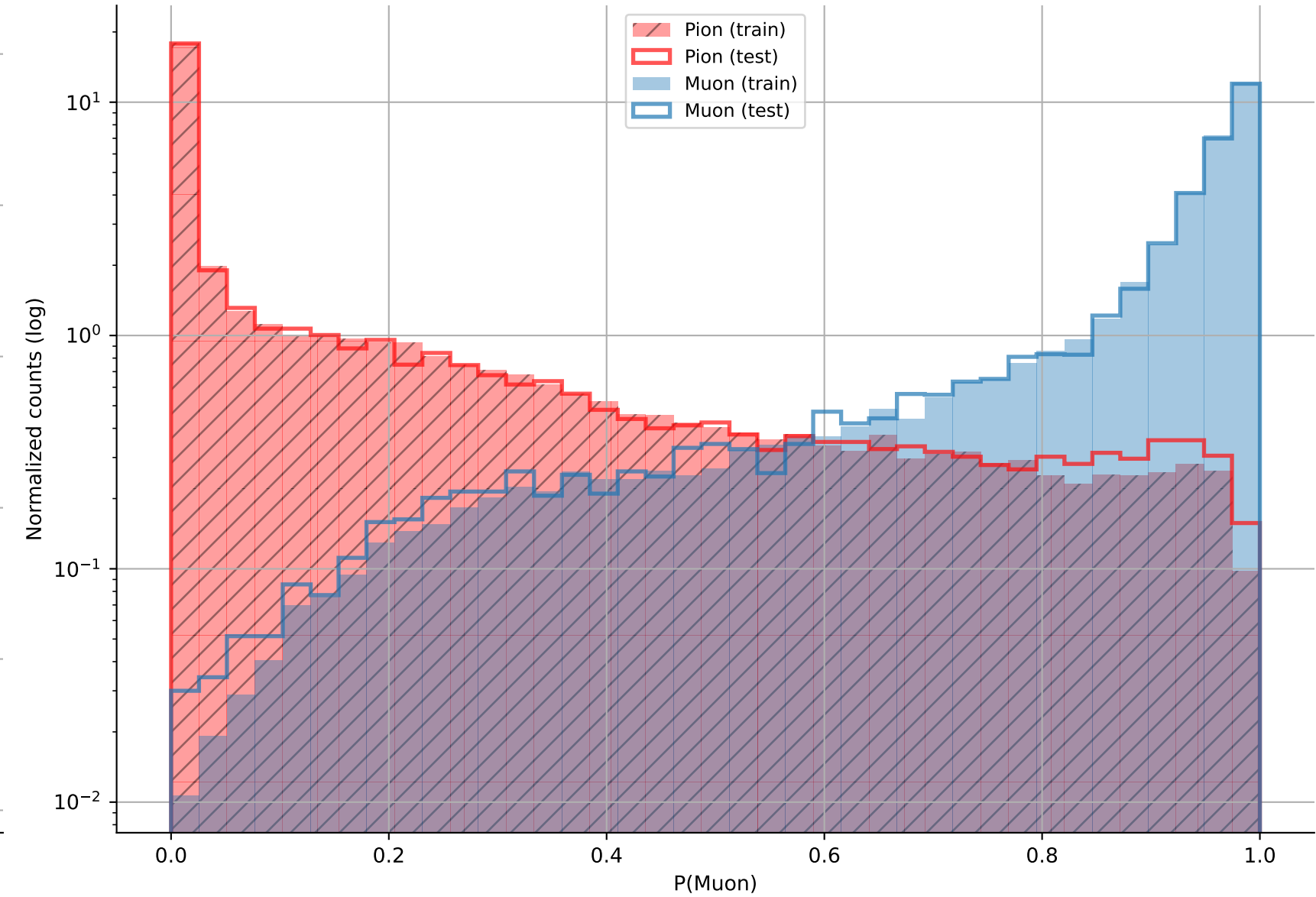


XGBoost Training Diagnostics

AUC vs Boosting Rounds

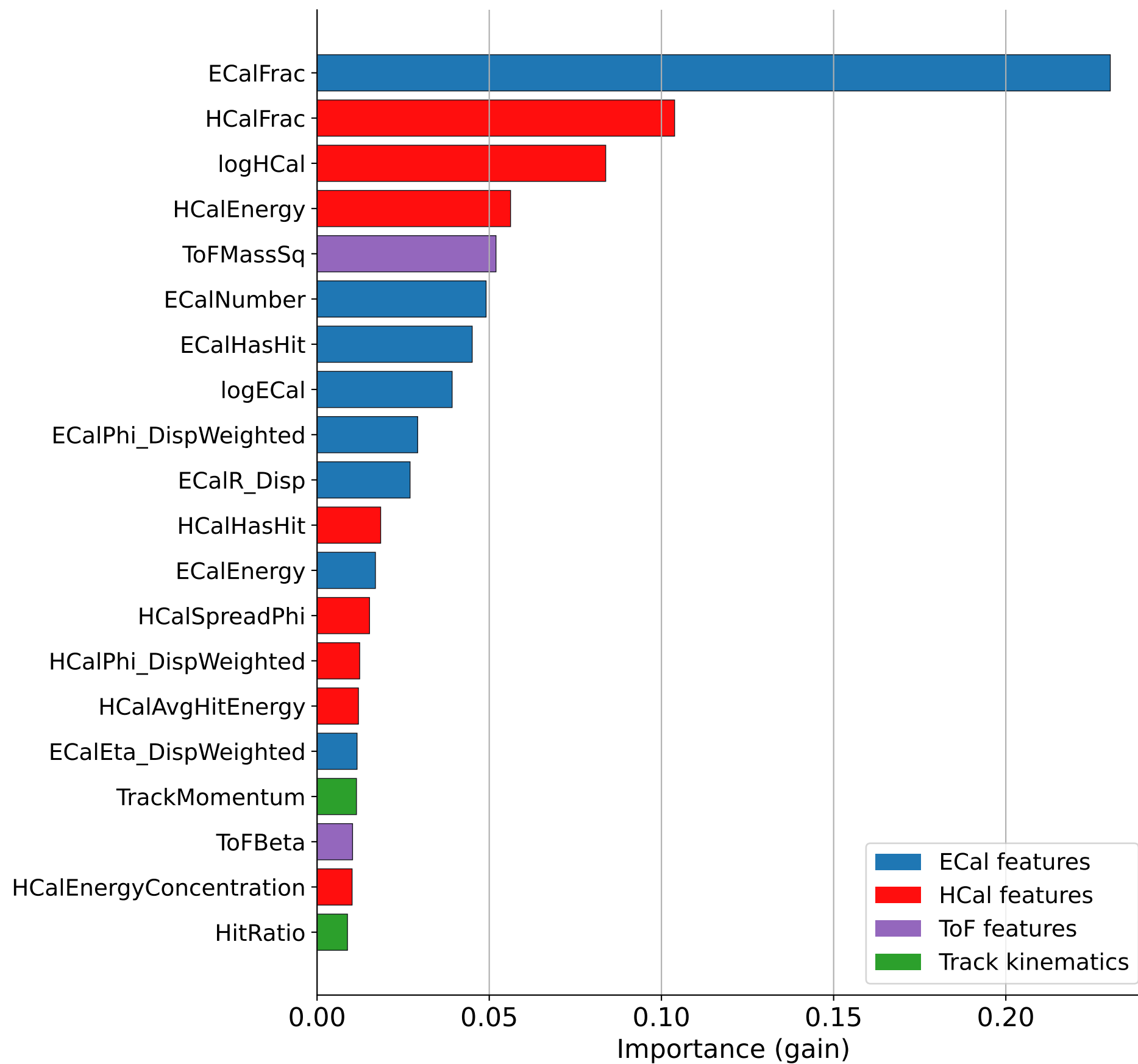


Overtraining Check (Train vs Test)

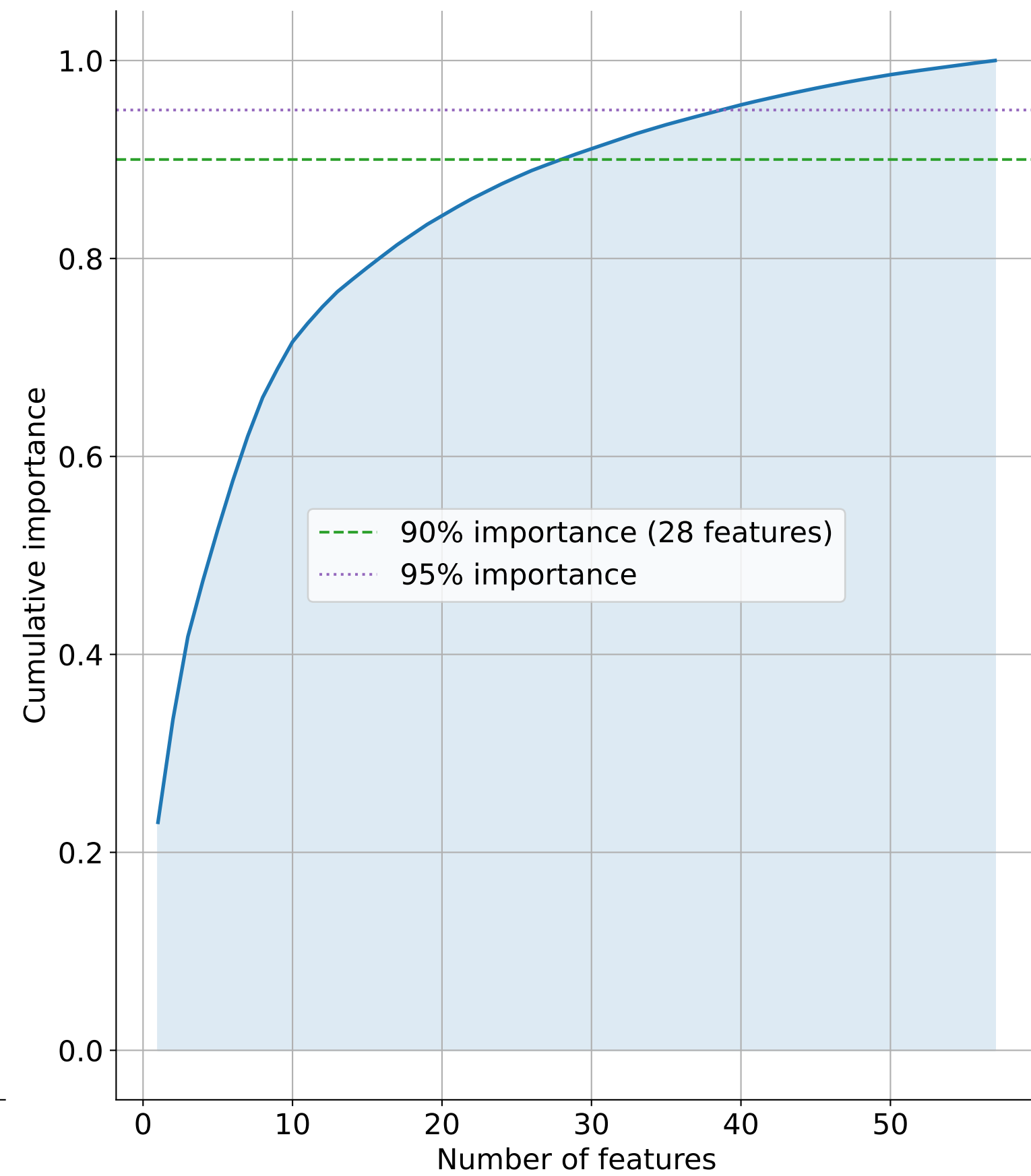


Feature Importances — XGBoost

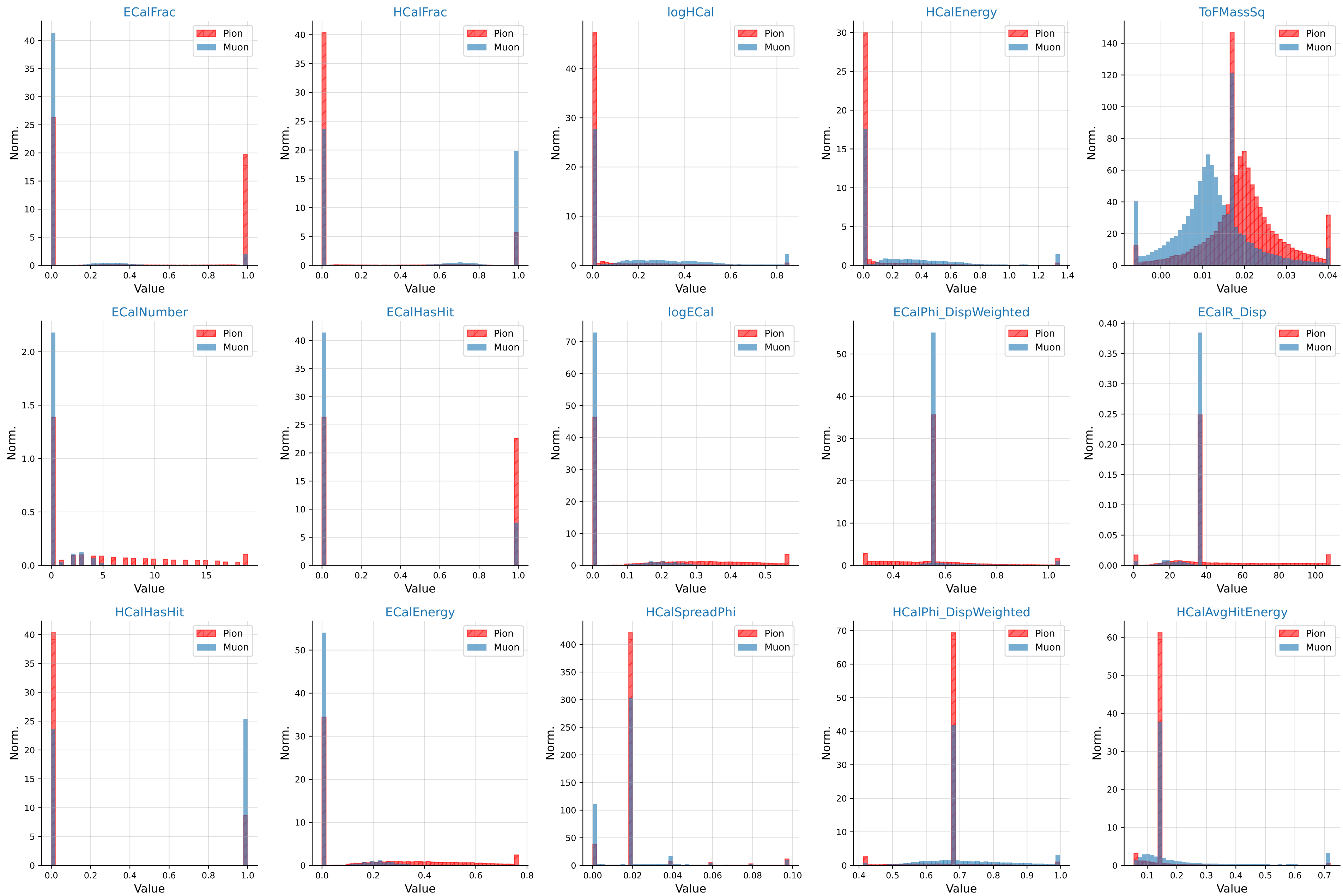
Top 20 Features



Cumulative Feature Importance

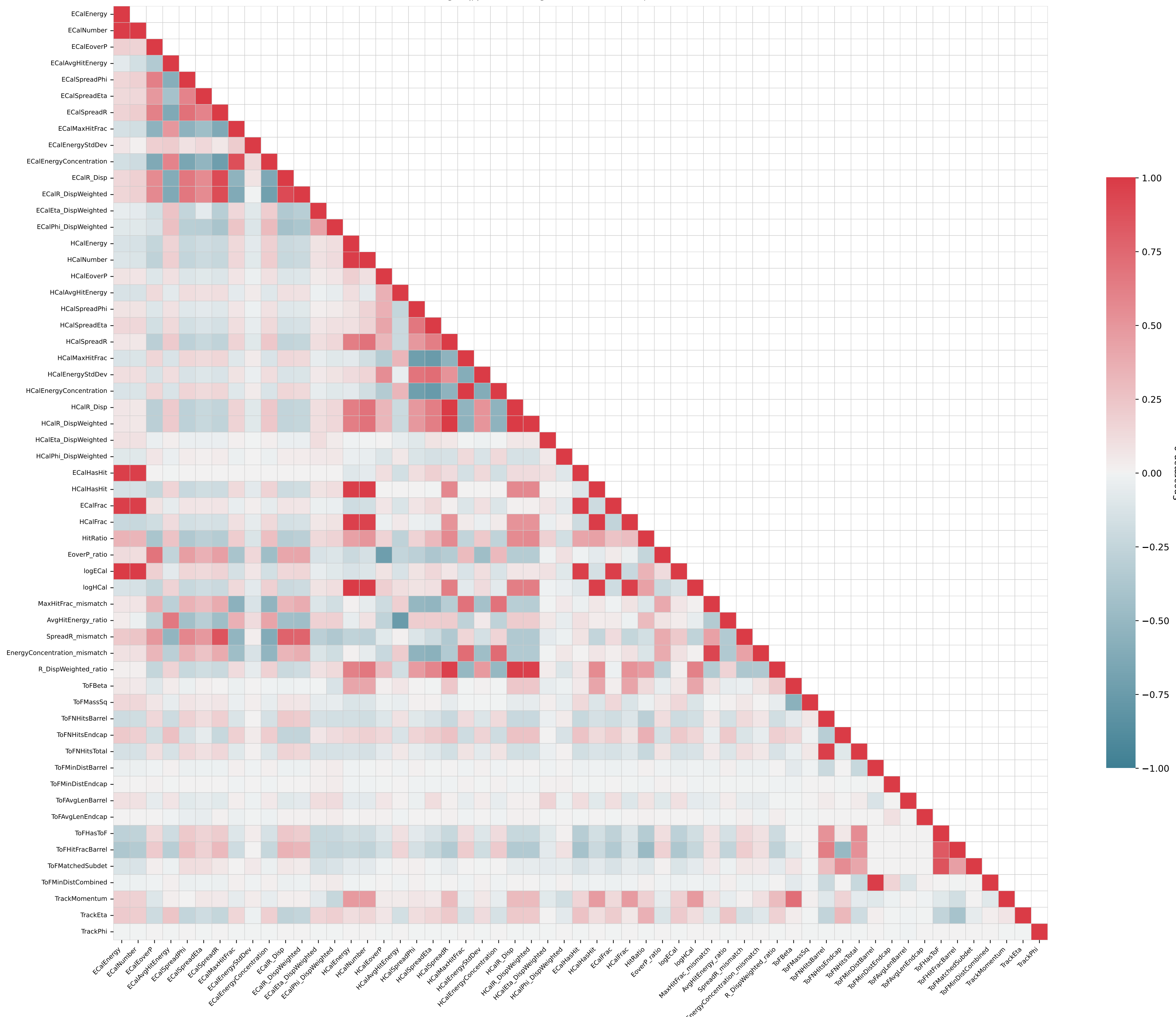


Key Physics Variable Distributions: Muon vs Pion [full dataset before splitting]

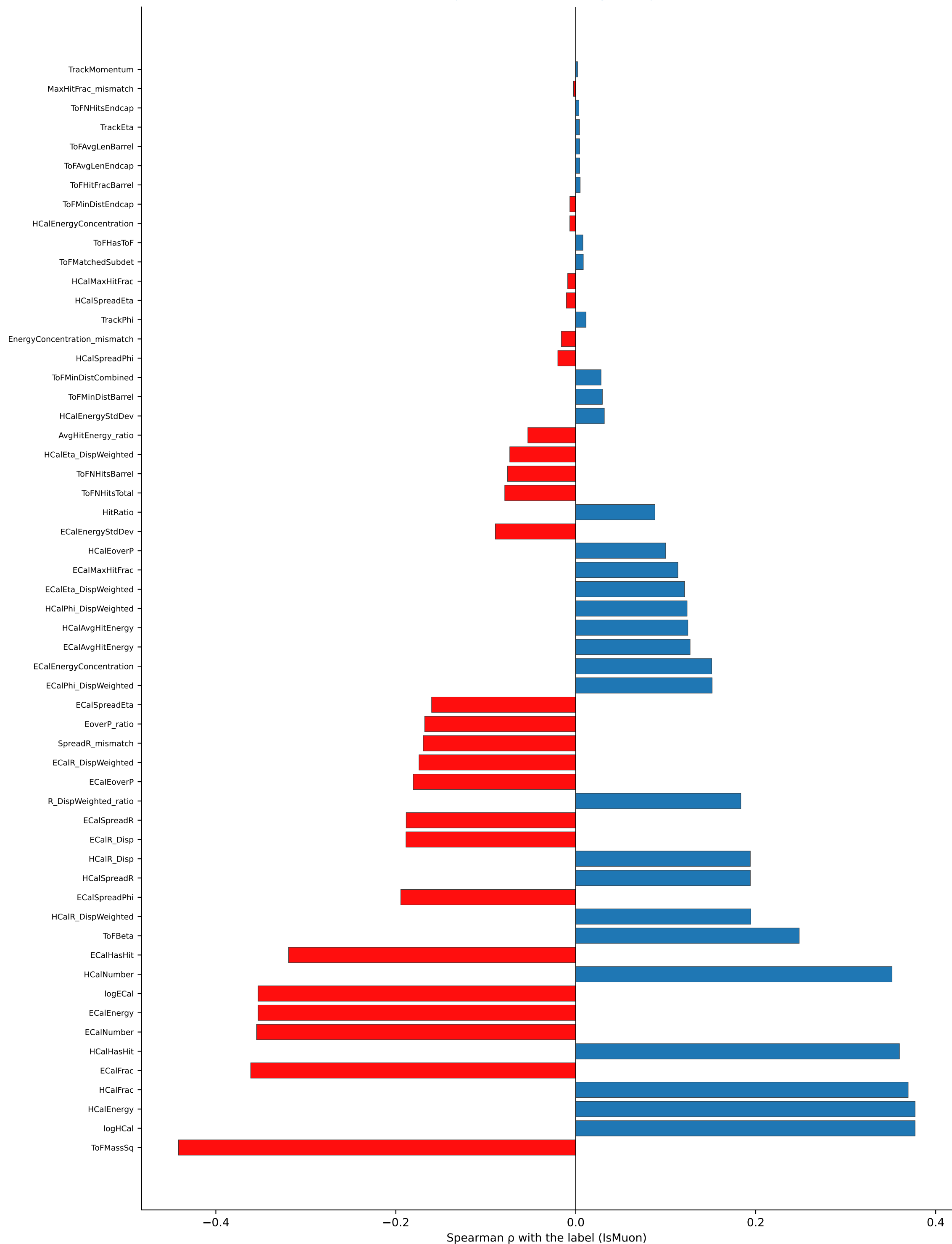


Spearman Correlation Matrix — All Features
(values from the full dataset)

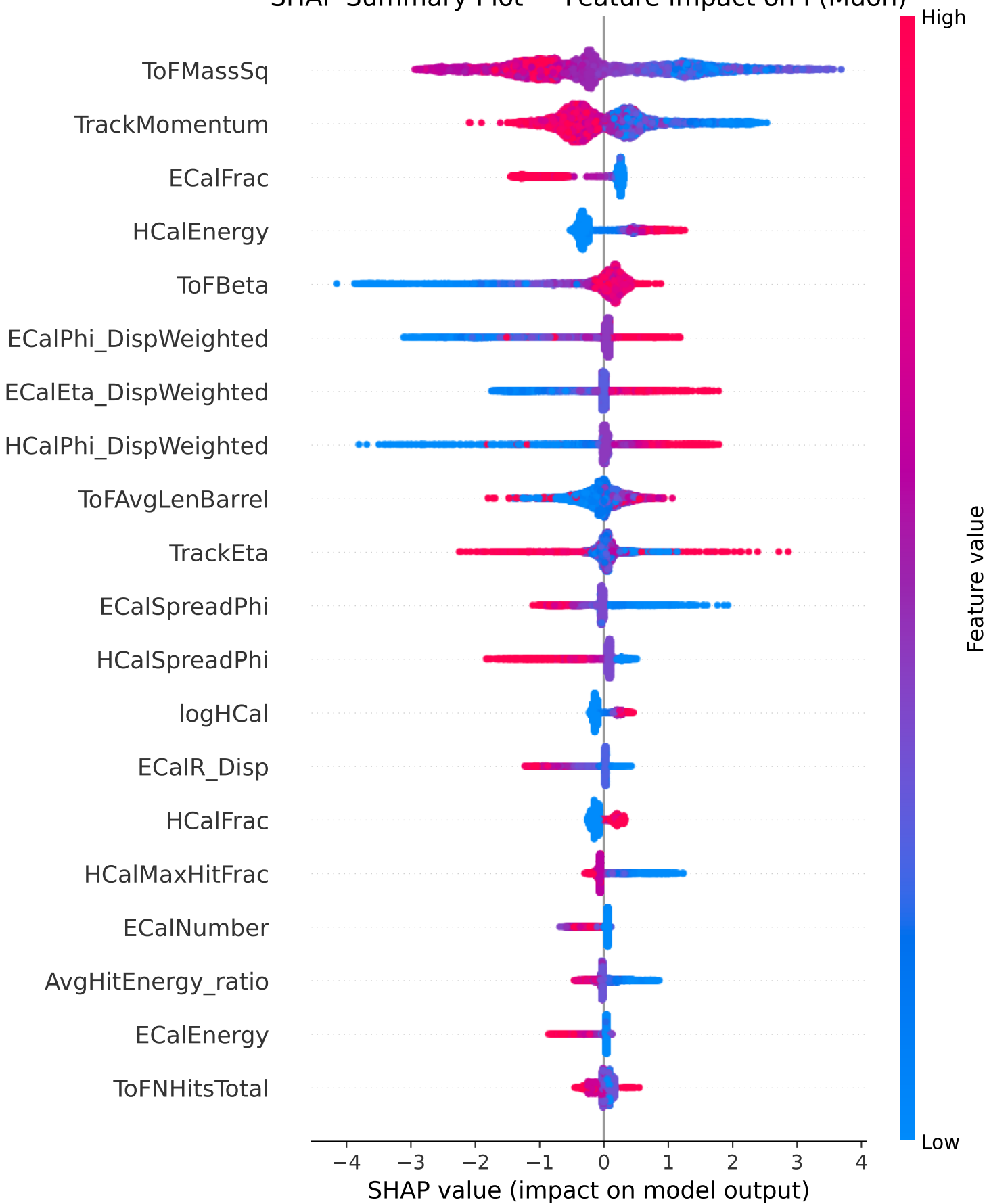
Higher $|\rho|$ means a stronger monotonic relationship between features



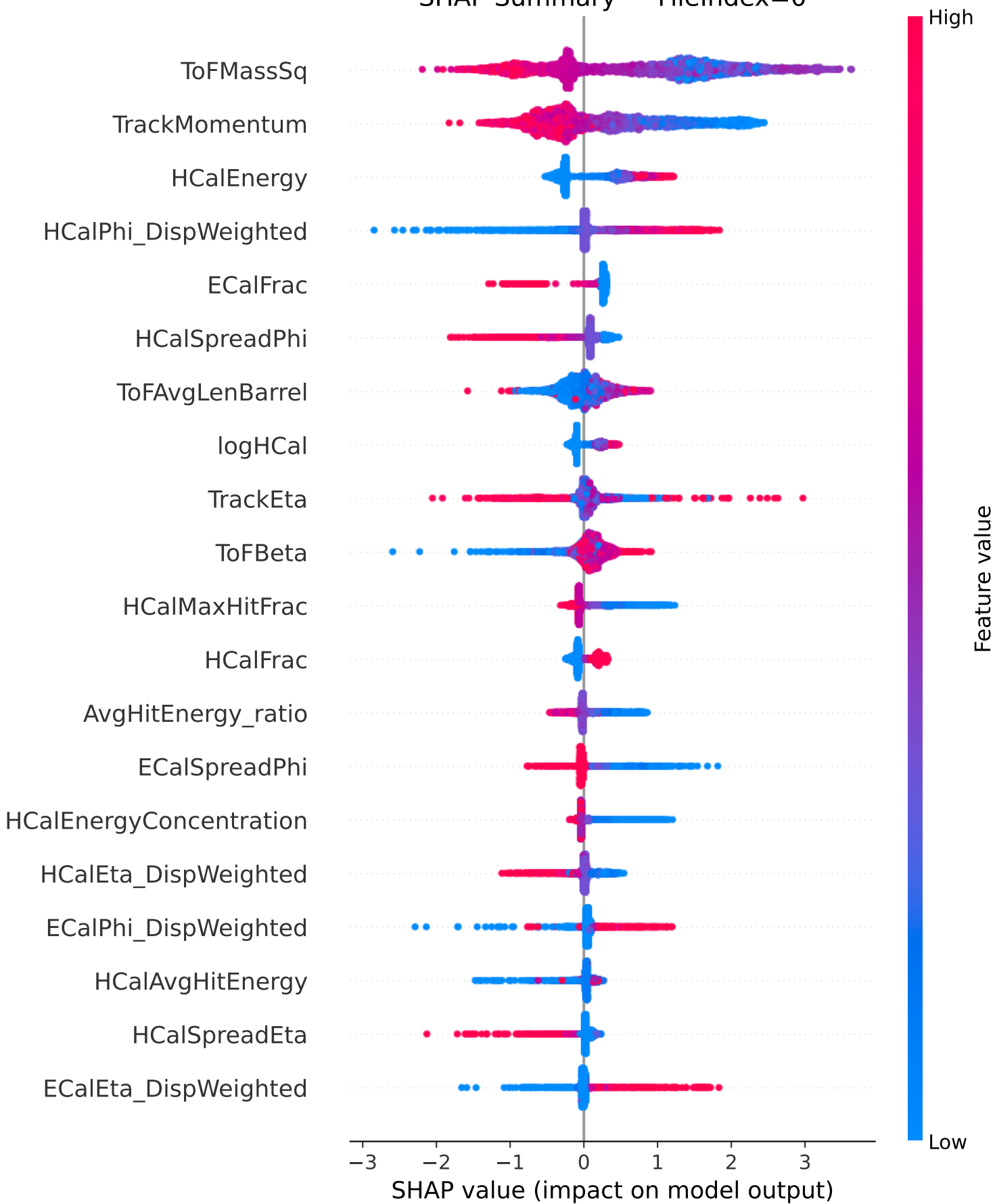
Feature correlation with muon/pion class
Blue = positive (→ muon), Red = negative (→ pion)



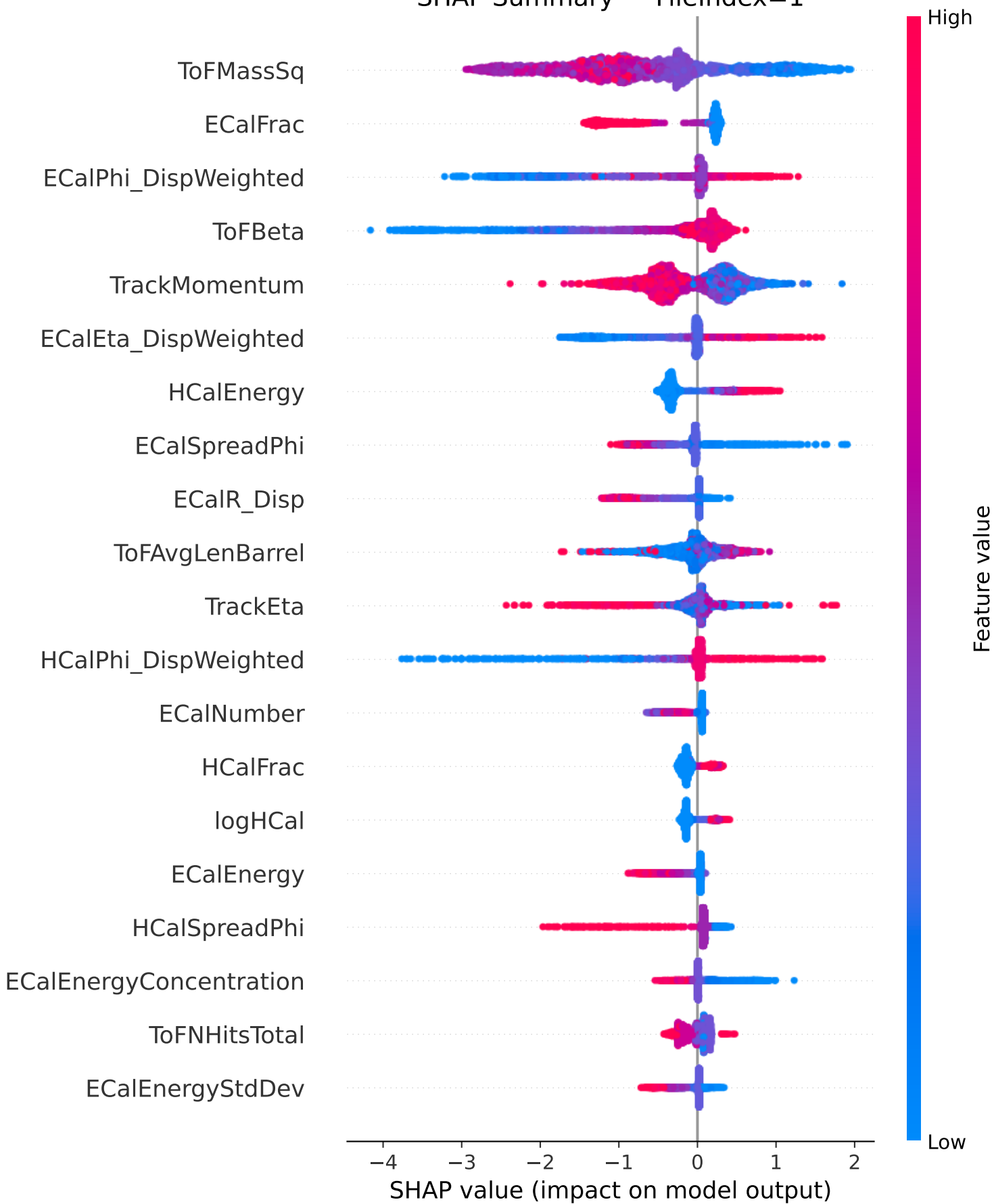
SHAP Summary Plot — Feature Impact on P(Muon)



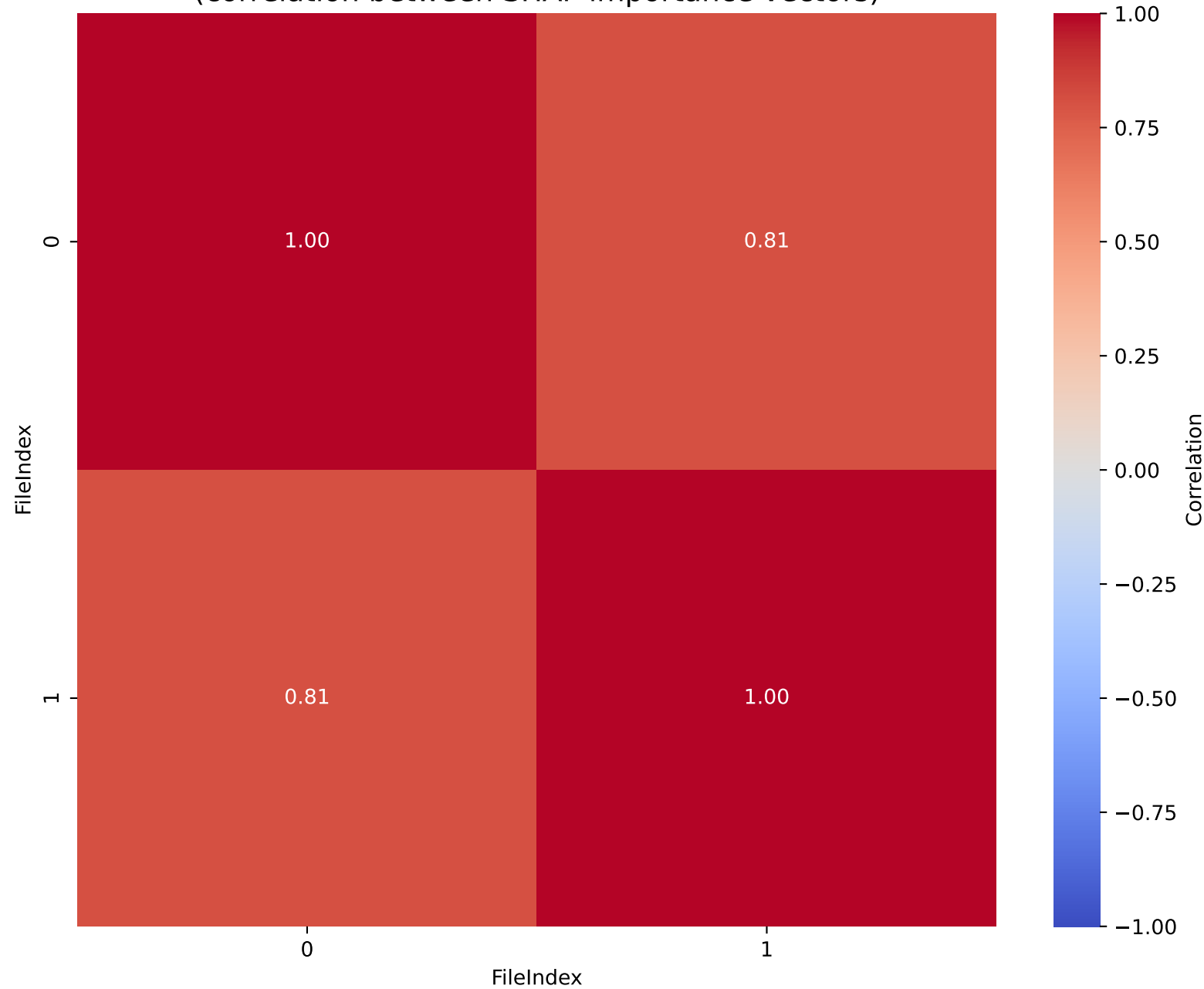
SHAP Summary — FileIndex=0



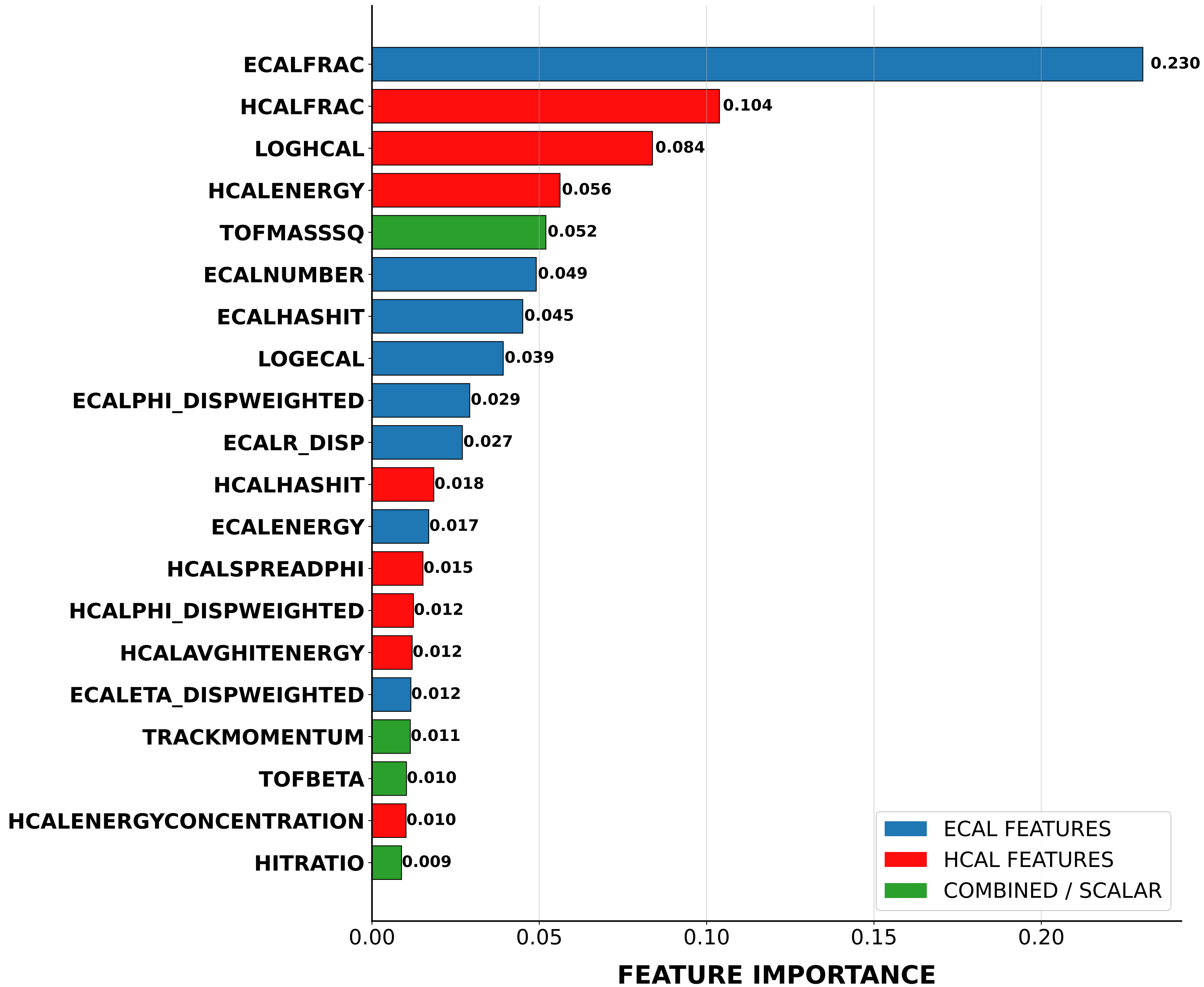
SHAP Summary — FileIndex=1



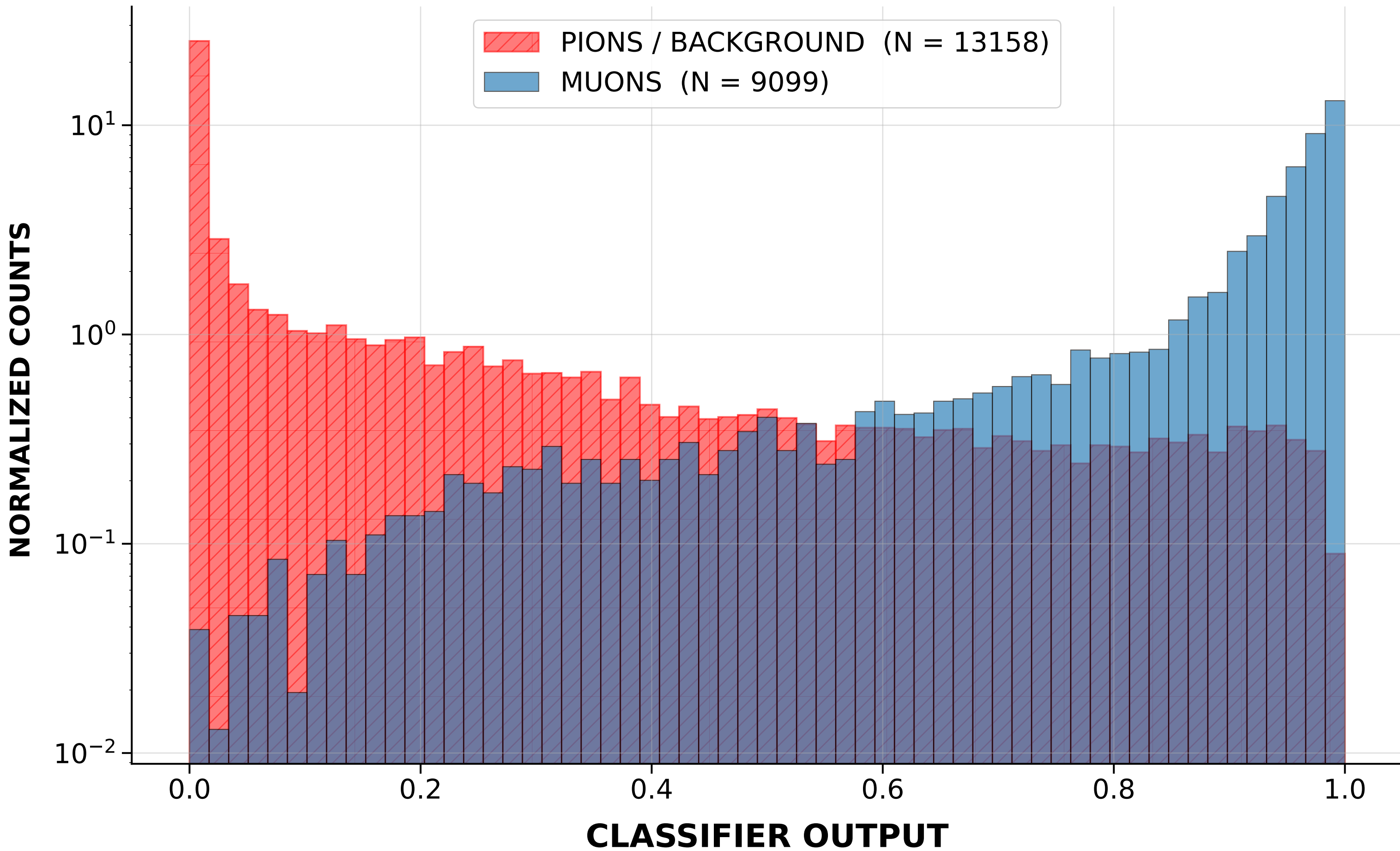
SHAP Similarity Matrix — FileIndex vs FileIndex
(correlation between SHAP importance vectors)



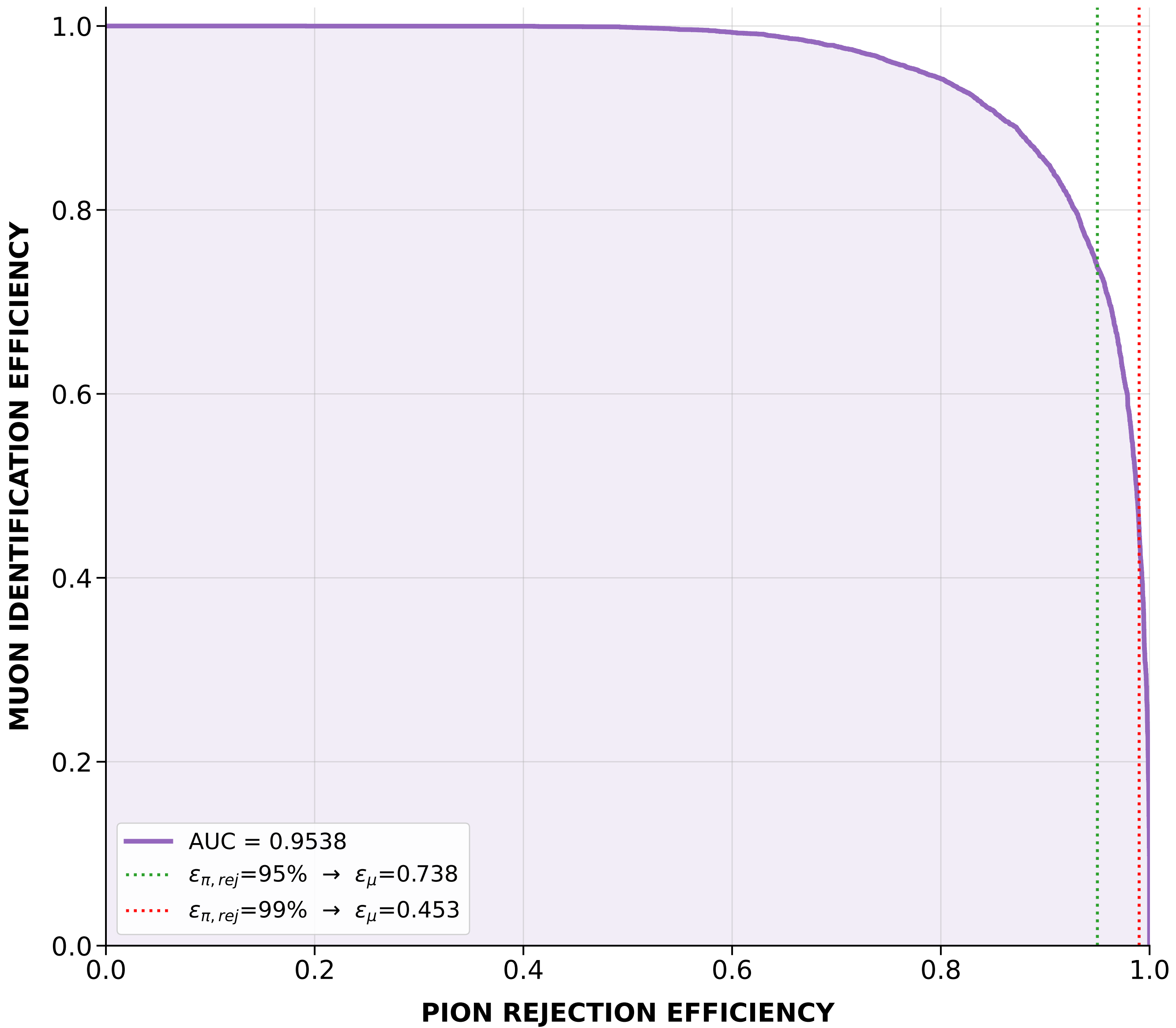
FEATURE IMPORTANCE — TOP 20



CLASSIFIER SCORE DISTRIBUTION MUONS vs PIONS

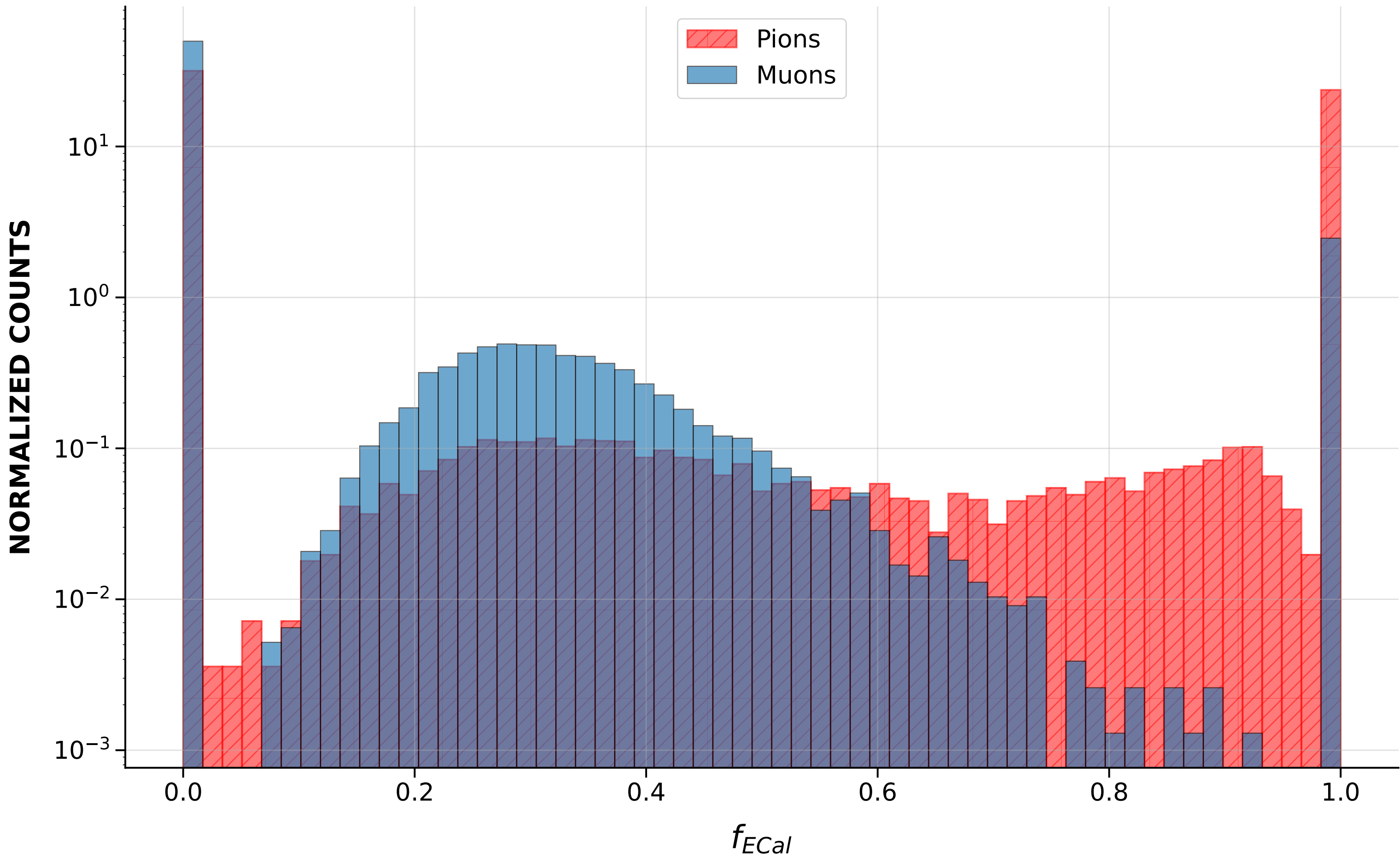


ROC CURVE: ε_μ VS $\varepsilon_{\pi, \text{rej}}$



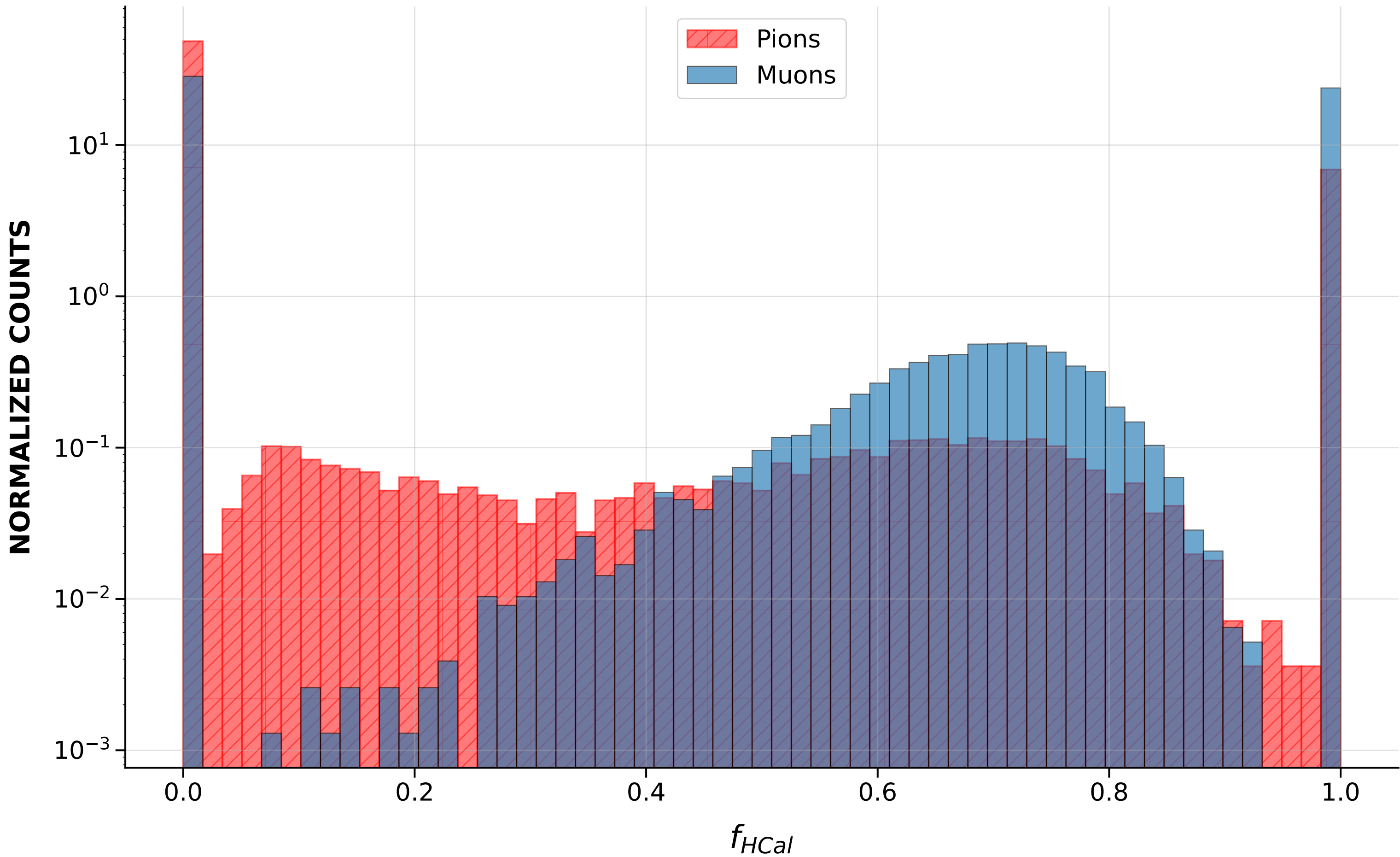
ENGINEERED CALORIMETER FEATURE

ECalFrac



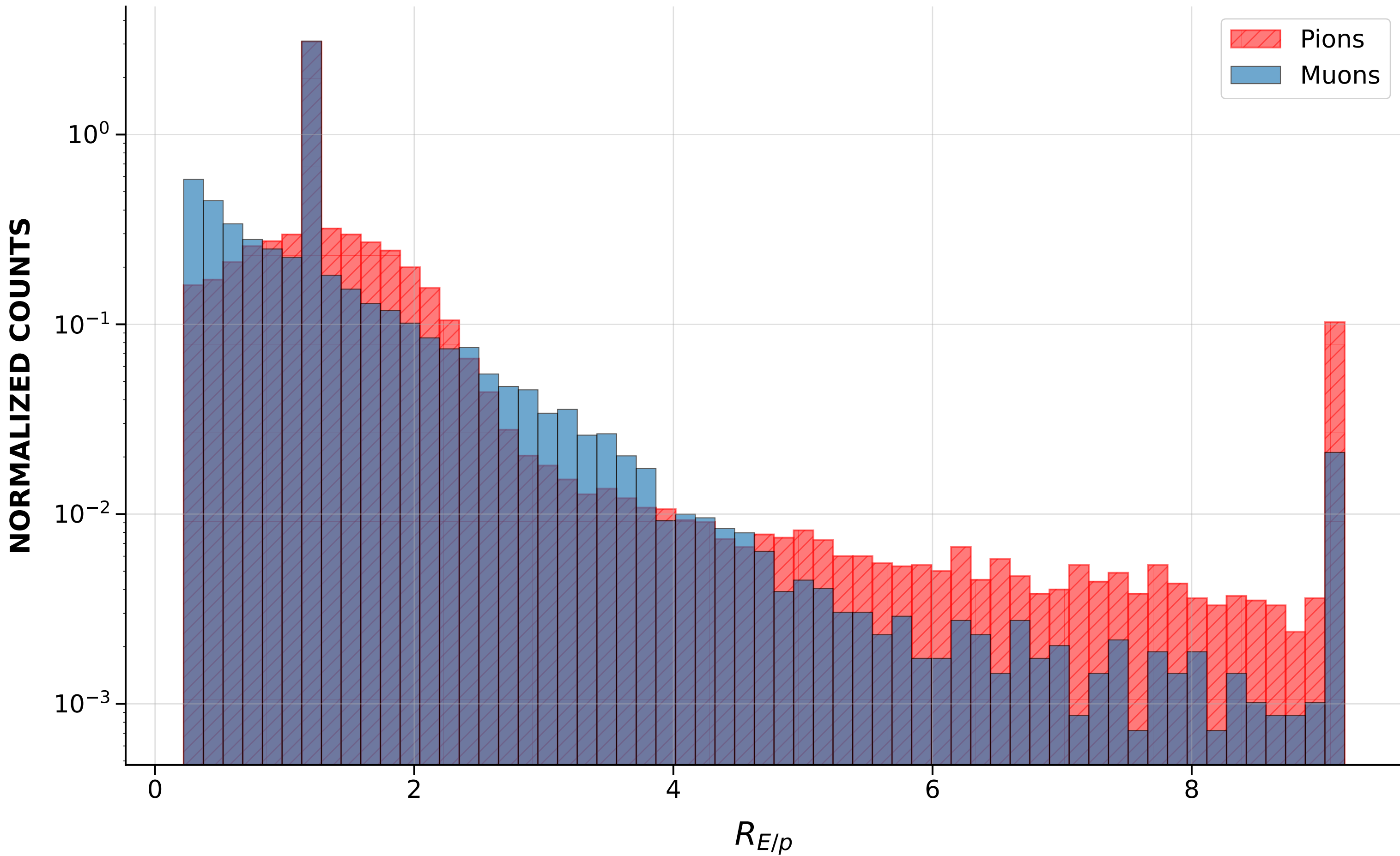
ENGINEERED CALORIMETER FEATURE

HCalFrac



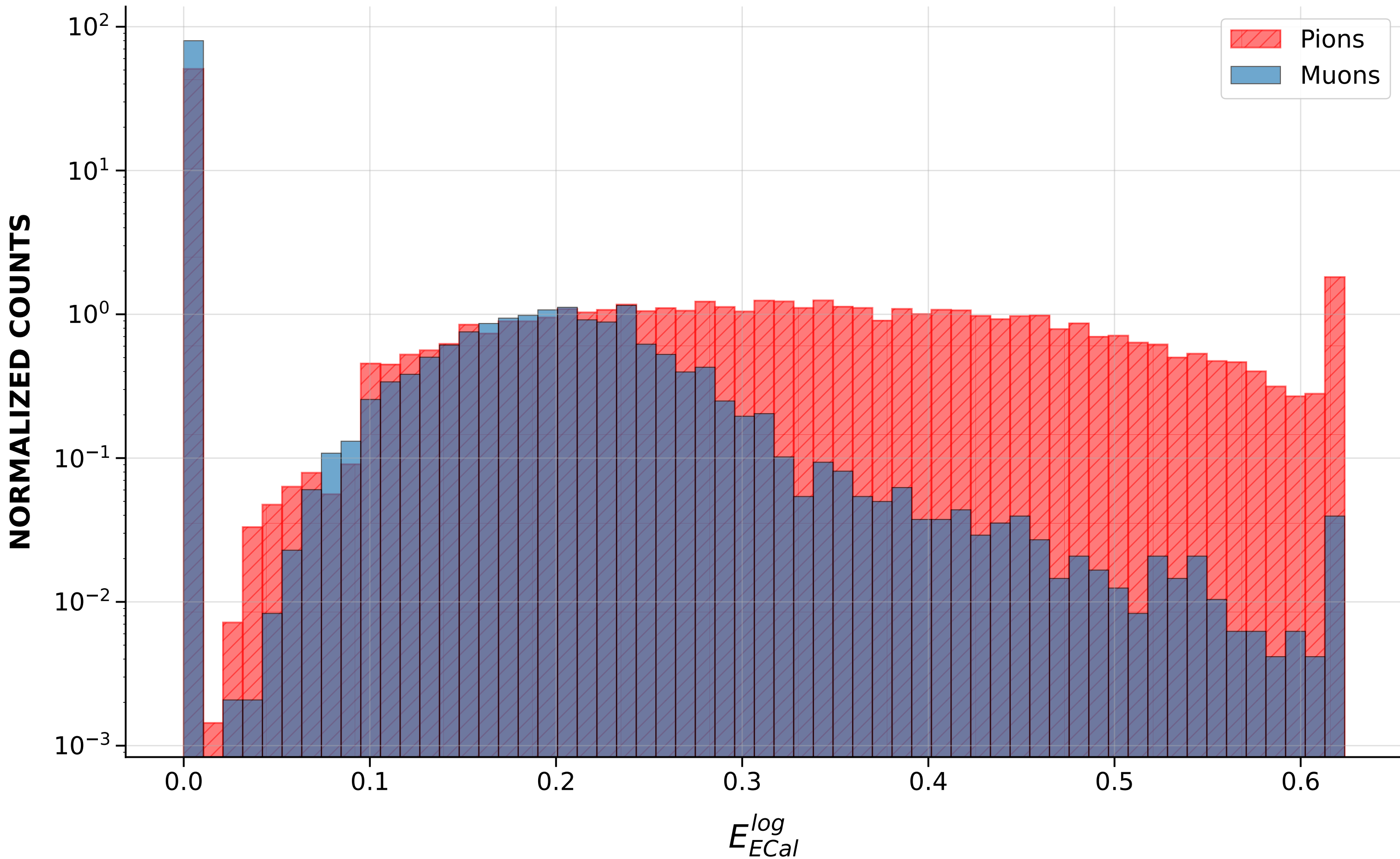
ENGINEERED CALORIMETER FEATURE

EoverP_ratio



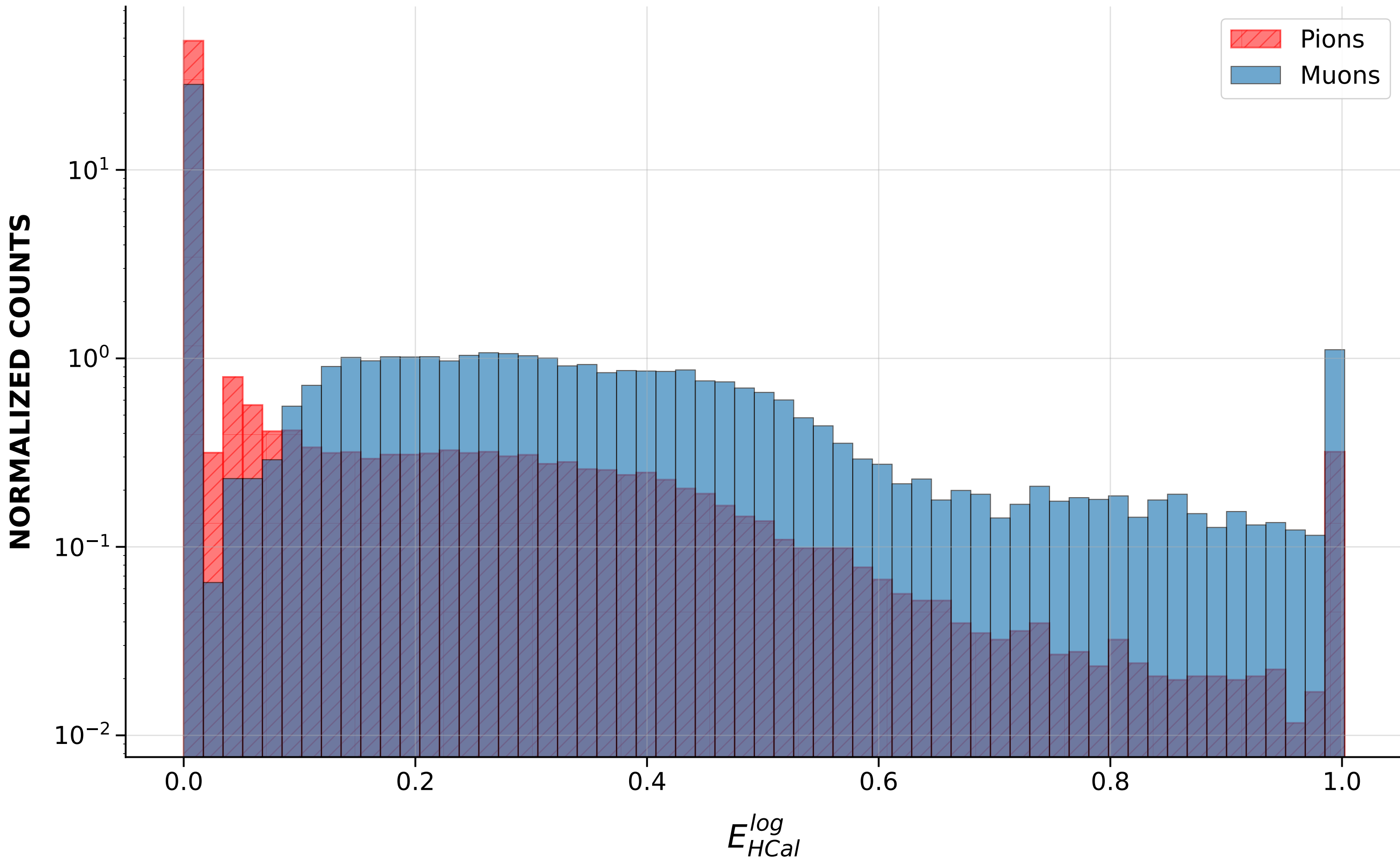
ENGINEERED CALORIMETER FEATURE

$\log E_{Cal}$



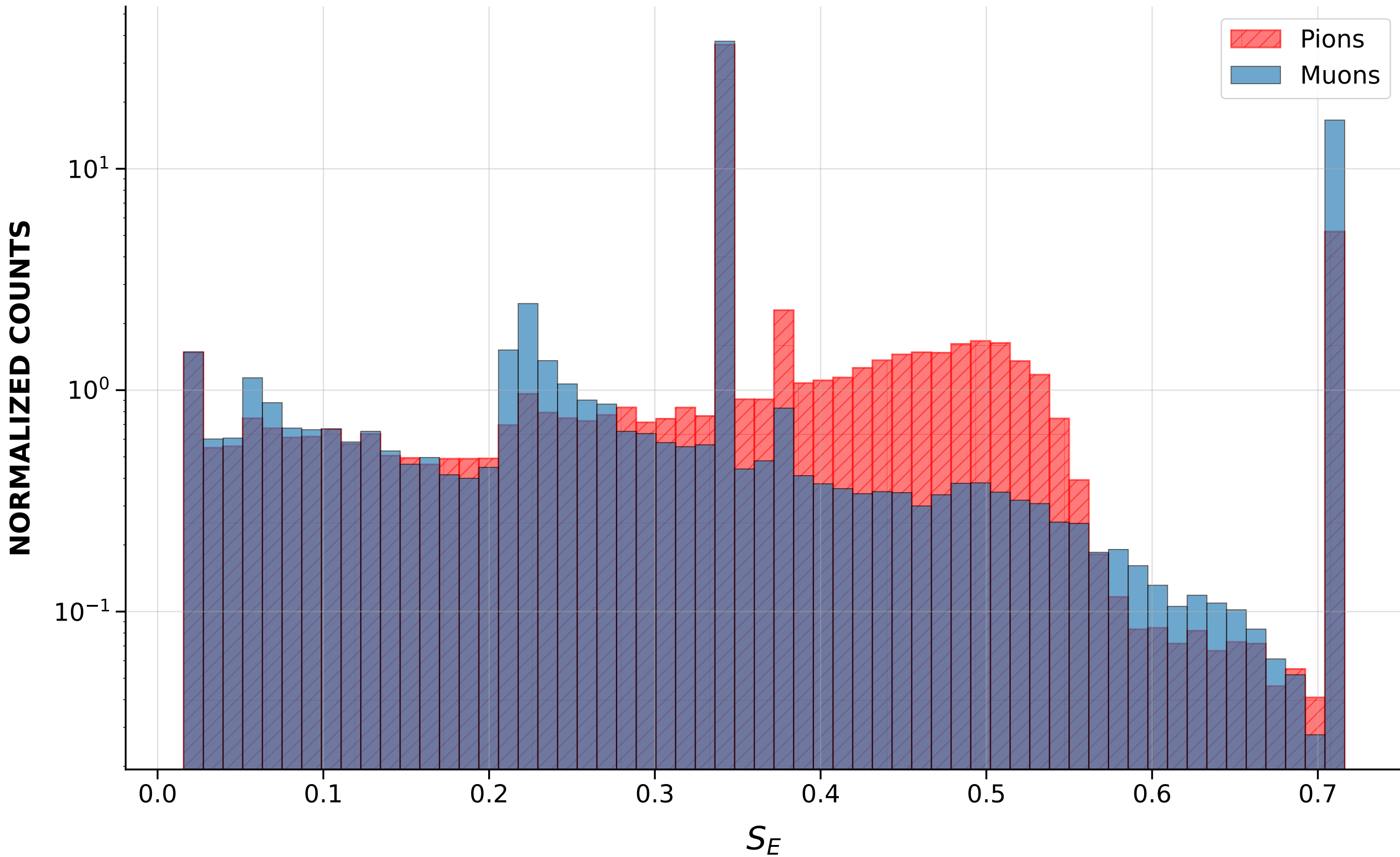
ENGINEERED CALORIMETER FEATURE

$\log H_{Cal}$



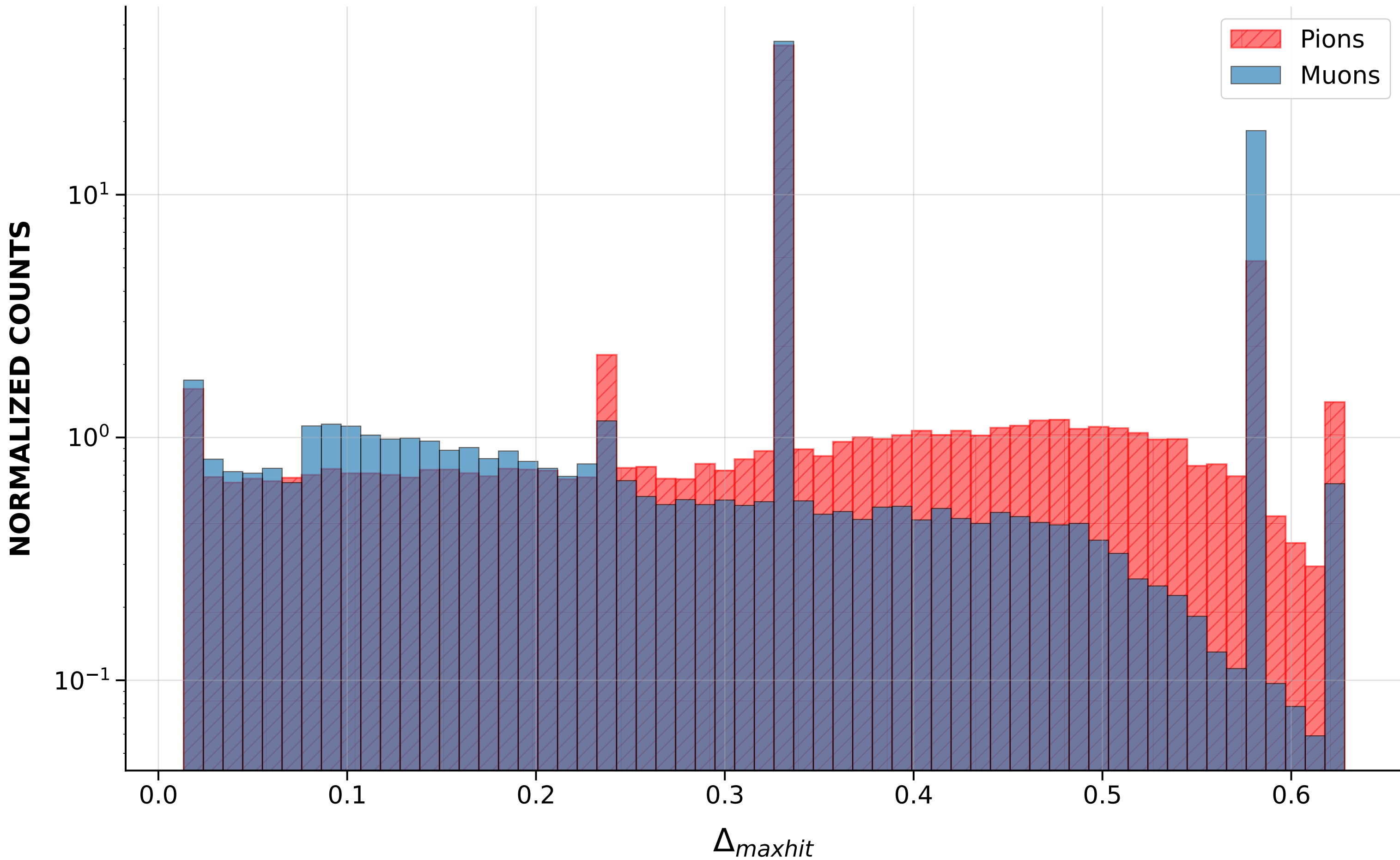
ENGINEERED CALORIMETER FEATURE

EnergyConcentration_mismatch



ENGINEERED CALORIMETER FEATURE

MaxHitFrac_mismatch



ENGINEERED CALORIMETER FEATURE

HitRatio

